

Reference excerpt 01-033

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the z -axes at instant t_{QP} . pressure in the IJV , measured in G, can be obtained simultaneously with the acquisition of the ECG trace [Sisini et al.(2016)]. On this trace, that reports both JVP and ECG waves, the time interval τ_{QP} between t_P Q and t_a is given by: $\tau_{QP} = t_a - t_{QP} = IG \cdot c$ (4) where t_a is the instant corresponding the a wave. The parameter τ_{QP} is the time interval between the instant when the pressure is maximum at point G (instant t_a) and the instant when the pressure is maximum at point RA (instant t_{QP}). METHODS The applicability of the relationship expressed in Eq. 4, can be tested by comparing the instantaneous velocity ($w(t)$) of the blood in the IJV calculated according to the equation shown above, with the instantaneous blood velocity ($w_D(t)$) measured with a Doppler scanner. However, the data and the results presented here are for illustration only and are not to be considered the result of a scientific study. Pressure waves velocity calculation The delay τ_{QP} is measured over the JVP+ECG trace as shown in Fig. 2. The distance IG is measured on the volunteers chest. The velocity c is the ratio between IG and τ_{QP} . Compliance calculation From the Moens-Korteweg equation we obtain the compliance per unit length: $C = CSA \cdot c^2$ (5) where CSA_x is the CSA of the IJV measured in G at the x wave. Pressure gradient calculation The pressure inside the IJV is calculated from the